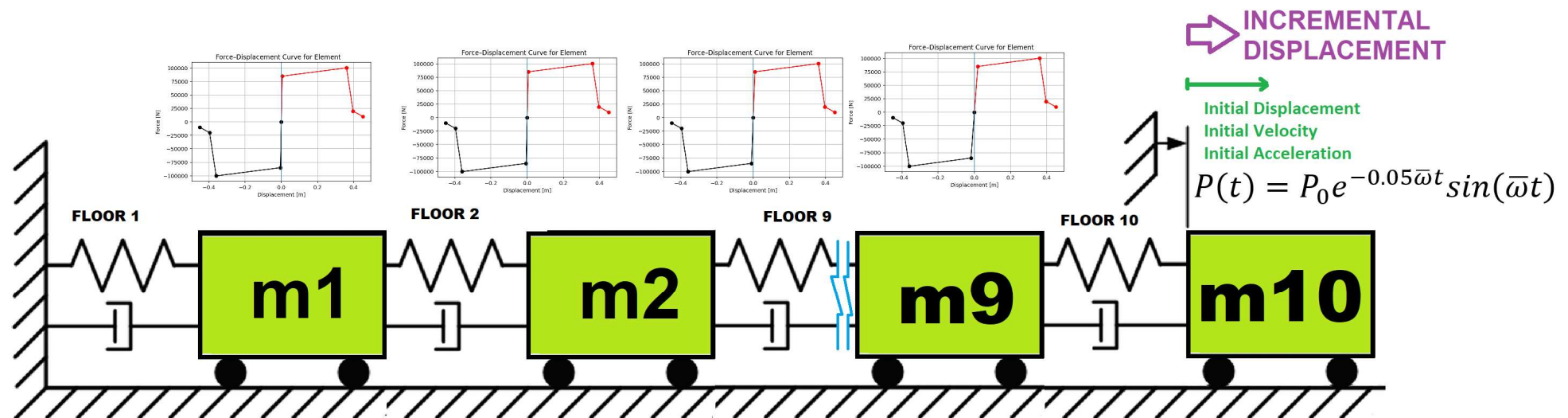


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF THE OPTIMUM ELASTIC STIFFNESS OF EACH COLUMN BASED ON THE STRUCTURAL PERIOD VARIATIONS THROUGH CONSTANT MASS USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

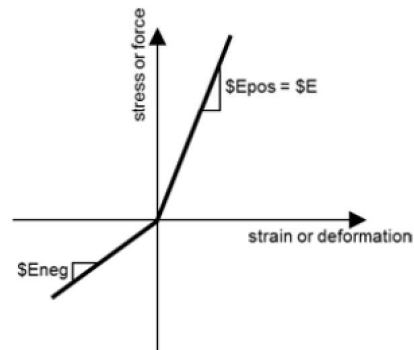
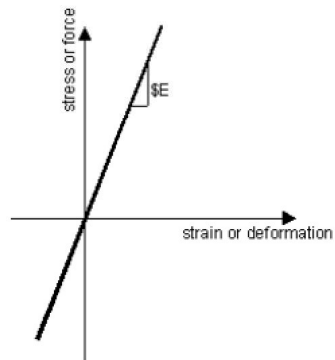
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

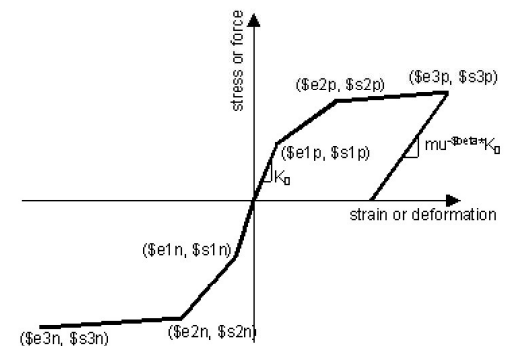
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

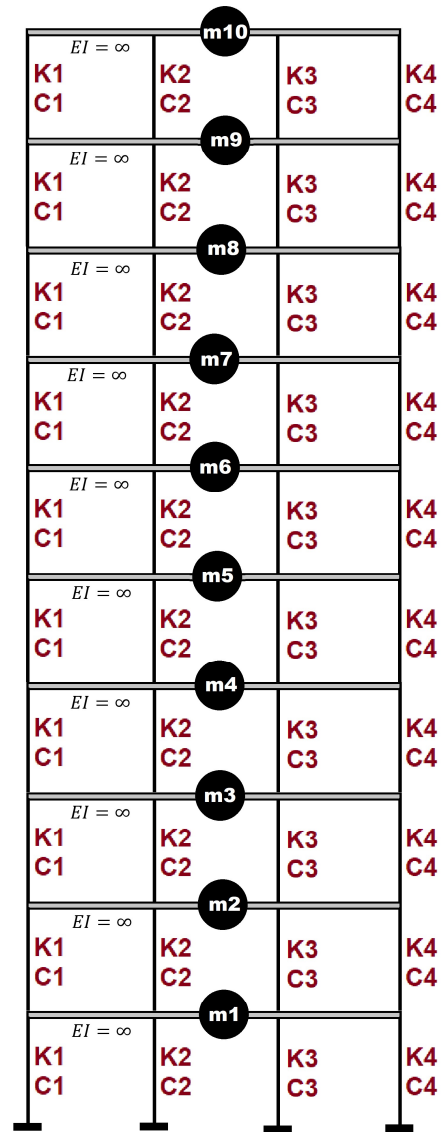
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



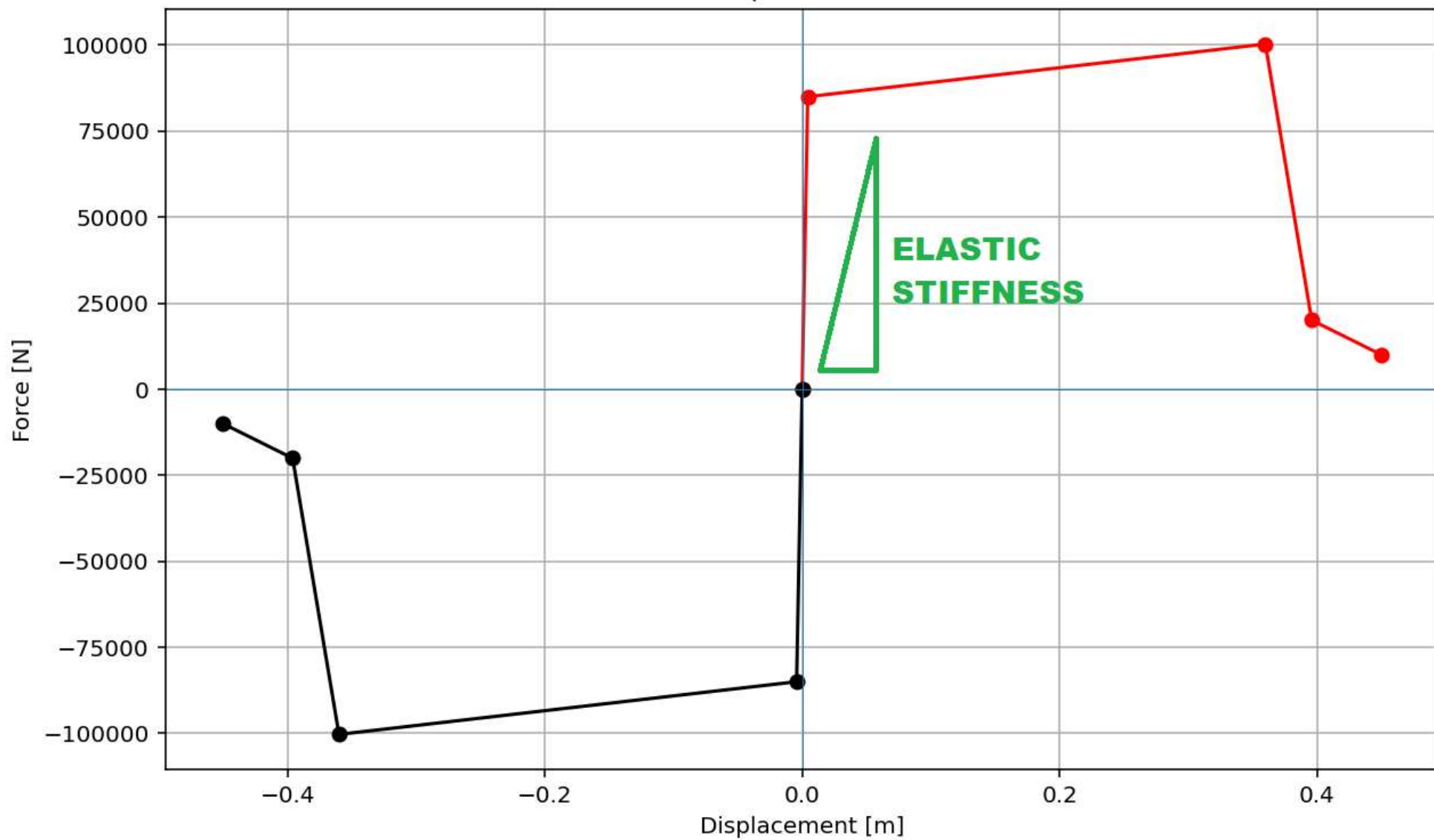
ELASTIC MATERIAL



INELASTIC MATERIAL



Force-Displacement Curve



**OPTIMIZATION
PROBLEM FOR
STRUCTURAL PERIOD
BY CHANGING EACH
COLUMN ELASTIC
STIFFNESS**

\\?C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_A..._OPTIMIZATION_PERIOD_MDOF_10_FLOORS_04_COLUMNS.py

KE_OPTIMIZATION_PE...OORS_04_COLUMNS.py X

```

1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF THE OPTIMUM ELASTIC STIFFNESS OF EACH COLUMN BASED ON THE STRUCTURAL PERIOD
4 # VARIATIONS THROUGH CONSTANT MASS USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
5 #-----
6 # IN THIS ANALYSIS MASS IS CONSTANT BUT DEMAND VALUE (STRUCTURAL PERIOD) IS NOT CONSTANT
7 #-----
8 # EQUIVALENT VISCOUS DAMPING RATIO:  $\xi_{eq} = 100 * E_d / (4 * \pi * E_s)$ 
9 #-----
10 # ENERGY DISSIPATION CAPACITY INDEX =  $100 * E_d(\text{earthquake}) / E_d(\text{cyclic displacement})$ 
11 #-----
12 # EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAL
13 #-----
14 #  $P(t) = P_0 \sin(\omega t)$ 
15 #  $P(t) = P_0 \exp(-0.05\omega t) \sin(\omega t)$ 
16 #-----
17 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVALUATION
18 # OF ENERGY DISSIPATION CAPACITY INDEX
19 #-----
20 # MARKOV CHAIN FOR EVALUTIONS OF PROBABILITY OF FAILURE
21 #-----
22 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)
23 # EMAIL: salar.d.ghashghaei@gmail.com
24 #####
25 """
26 Nonlinear Seismic Performance Assessment of a MDOF:
27 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Earthquake L
28
29 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a MDOF
30 for performance-based earthquake engineering. The 2D model incorporates both material nonlinearity
31 (elastic-perfectly plastic or hysteretic steel with strain hardening)
32 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
33 and large displacements-essential for collapse assessment.
34

```

...older\OPTIMIZATION_KE_40_PERIOD_MDOF_10_FLOORS_04_COLUMNS+

Console 1/A X

1031 2031
1032 2032
1033 2033
1034 2034
1035 2035
1036 2036
1037 2037
1038 2038
1039 2039

lambda	omega	period	frequency
2.467e+00	1.5708	4.0000	0.2500
2.180e+01	4.6695	1.3456	0.7432
5.838e+01	7.6409	0.8223	1.2161
1.082e+02	10.4039	0.6039	1.6558
1.660e+02	12.8830	0.4877	2.0504

Fmax: -1.3708056911809763e-10
DF: -1.3707168733390063e-07
DX: 1.0043478542594068e-07
IT: 423 - RESIDUAL: 1.0043478542594068e-07 X: 14589660.98997329

Structural Period: 4.0000
Optimum X (COL. ELASTIC STIFFNESS): 14589660.9900
Iteration Counts: 423
Convergence Residual: 1.0043478543e-07

Total time (s): 4.6250

IPython Console Plots Variable Explorer Help Debugger History Files

Fmax: -1.3708056911809763e-10

DF: -1.3707168733390063e-07

DX: 1.0043478542594068e-07

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Optimum X (COL. ELASTIC STIFFNESS): 14589660.9900

Iteration Counts: 423

Convergence Residual: 1.0043478543e-07

Total time (s): 4.6250

STRUCTURAL PERIOD [s] and OPTIMUM ELASTIC COL. STIFFNESS [N/m] DURING 40 TIMES PERIOD ANALYSIS BY CHANGING THE TARGETED STRUCTURAL PERIOD: 4.000 [s]

